

5th week

1.	$\frac{dy}{dx} = 2x+3$ $\Rightarrow dy = (2x+3)dx$ $\Rightarrow \int dy = \int (2x+3)dx$ $y = x^2+3x+C$	$\Rightarrow dy = (2x+3)dx$ $\int dy = \int (2x+3)dx$ $y = x^2+3x+C$	$\Rightarrow dy = 2x+3 dx$ $\int dy = \int (2x+3) dx$ $y = x^2+3x+C$
2.	$\frac{dy}{dx} = 2x+3, y(1)=2$ $y = x^2+3x+C$ $\because y(1)=2$ $\Rightarrow 2 = 4+C \Rightarrow C = -2$ $\Rightarrow y = x^2+3x-2$	$y = x^2+3x+C$ $\because y(1)=2$ $\Rightarrow 2 = 4+C \Rightarrow C = -2$ $\Rightarrow y = x^2+3x-2$	$y = x^2+3x+C$ $\because y(1)=2$ $\Rightarrow 2 = 4+C \Rightarrow C = -2$ $\Rightarrow y = x^2+3x-2$
3.	$\frac{dy}{dx} = \frac{2x+3}{3y^2}, y(0)=2$ $\int 3y^2 dy = \int (2x+3)dx$ $y^3 = x^2+3x+C$ $\because y(0)=2$ $\Rightarrow 8 = C \Rightarrow y = \sqrt[3]{x^2+3x+8}$	$\int 3y^2 dy = \int (2x+3)dx$ $\Rightarrow y^3 = x^2+3x+C$ $\because y(0)=2$ $\Rightarrow 8 = C \Rightarrow y = \sqrt[3]{x^2+3x+8}$	$\int 3y^2 dy = \int (2x+3)dx$ $\Rightarrow y^3 = x^2+3x+C$ $\because y(0)=2$ $\Rightarrow 8 = C \Rightarrow y = \sqrt[3]{x^2+3x+8}$
4.	$\frac{dy}{dx} = 2x+2xy, y(0)=2$ $\frac{dy}{dx} = 2x(1+y)$ $\Rightarrow \int \frac{1}{1+y} dy = \int 2x dx$ $\Rightarrow \ln 1+y = x^2+C_1$ $\Rightarrow 1+y = C_1 e^{x^2} \Rightarrow C_1 e^{x^2} - 1 = y$ $\Rightarrow \because y(0)=2 \Rightarrow C_1=3$ $\Rightarrow y = 3e^{x^2} - 1$	$\frac{dy}{dx} = 2x(1+y)$ $\Rightarrow \int \frac{1}{1+y} dy = \int 2x dx$ $\Rightarrow \ln 1+y = x^2+C_1$ $\Rightarrow 1+y = e^{x^2+C_1} = C_1 e^{x^2}$ $\Rightarrow y = C_1 e^{x^2} - 1 \Rightarrow \because y(0)=2$ $\Rightarrow C_1=3 \Rightarrow y = 3e^{x^2} - 1$	$\frac{dy}{dx} = 2x(1+y)$ $\Rightarrow \int \frac{1}{1+y} dy = \int 2x dx$ $\Rightarrow \ln 1+y = x^2+C_1$ $\Rightarrow 1+y = e^{x^2+C_1}$ $\Rightarrow 1+y = \pm C_1 e^{x^2} \Rightarrow y = C_1 e^{x^2} - 1 \because y(0)=2$ $\Rightarrow C_1=3 \Rightarrow y = 3e^{x^2} - 1$
5.	$\frac{dy}{dx} = \frac{e^{2x}+4x}{3y^2-\cos 2y}$ $\int (3y^2-\cos 2y) dy = \int (e^{2x}+4x) dx$ $\Rightarrow y^3 - \frac{1}{2} \sin 2y = \frac{1}{2} e^{2x} + 2x^2 + C$	$\frac{dy}{dx} = \frac{e^{2x}+4x}{3y^2-\cos 2y}$ $\Rightarrow \int (3y^2-\cos 2y) dy = \int (e^{2x}+4x) dx$ $\Rightarrow y^3 - \frac{1}{2} \sin 2y = \frac{1}{2} e^{2x} + 2x^2 + C$	

5th week

変化率
1.27% 1.25%
人数

6. If the numbers of covid-19 cases is 300,000,000 now and its rate is 1% per day, how many days will the cases be over 360,000,000? ($\ln(1.2) \approx 0.1823$)

Suppose that the numbers of covid-19 cases is y after x days

$$\text{Then } \frac{dy}{dx} = 0.01y$$

$$\int \frac{1}{y} dy = \int 0.01 dx$$

$$\Rightarrow \ln|y| = 0.01x + C_1$$

$$\Rightarrow |y| = C_1 e^{0.01x}$$

$$\Rightarrow y = C e^{0.01x}$$

$$\because y(0) = 300,000,000 \Rightarrow C = 300,000,000$$

$$\Rightarrow y = 300,000,000 e^{0.01x} \geq 360,000,000$$

$$\Rightarrow e^{0.01x} \geq 1.2 \Rightarrow 0.01x \geq \ln 1.2 \Rightarrow x \geq 18.23$$

$$\because x \in \mathbb{Z} \therefore x \geq 19$$

Suppose that the number of covid-19 cases is y after x days

$$\text{Then } \frac{dy}{dx} = 0.01y$$

$$\int \frac{1}{y} dy = \int 0.01 dx$$

$$\ln|y| = 0.01x + C_1$$

$$|y| = e^{0.01x + C_1}$$

$$y = C e^{0.01x}$$

$$\because y(0) = 300,000,000 \Rightarrow C = 300,000,000$$

$$\Rightarrow y = 300,000,000 e^{0.01x} \geq 360,000,000$$

$$\Rightarrow e^{0.01x} \geq 1.2 \Rightarrow 0.01x \geq \ln 1.2 \Rightarrow x \geq 18.23$$

$$\because x \in \mathbb{Z} \therefore x \geq 19$$

Suppose that the number of covid-19 cases is y after x days

$$\text{Then } \frac{dy}{dx} = 0.01y$$

$$\int \frac{1}{y} dy = \int 0.01 dx$$

$$\ln|y| = 0.01x + C_1$$

$$y = C e^{0.01x}$$

$$\because y(0) = 300,000,000 \Rightarrow C = 300,000,000$$

$$\Rightarrow y = 300,000,000 e^{0.01x} \geq 360,000,000$$

$$\Rightarrow e^{0.01x} \geq 1.2$$

$$\Rightarrow 0.01x \geq \ln 1.2 \Rightarrow x \geq 18.23$$

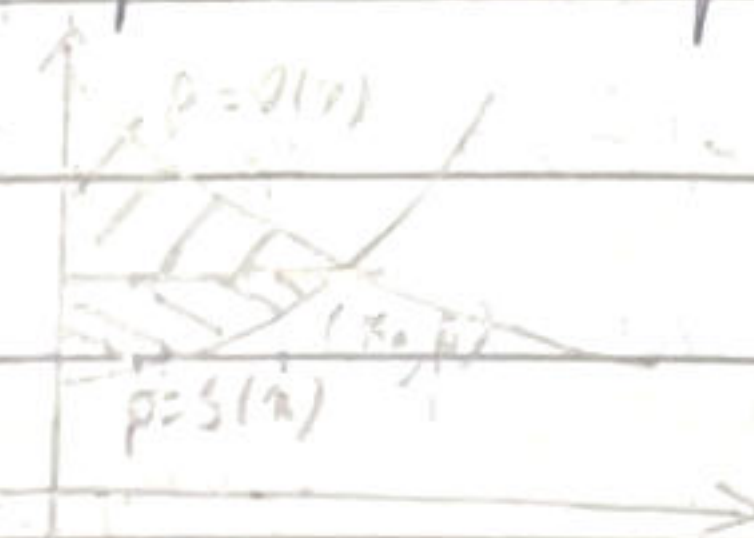
$$\because x \in \mathbb{Z} \therefore x \geq 19$$

7. The demand of a product is $D(x) = 1200 - 4x - 3x^2$ and production function is $S(x) = 540 + 2x + 3x^2$. Find the consumer surplus and the producer surplus.

Let the equilibrium point be $(x_0, \bar{p})$

$$\begin{cases} p = 1200 - 4x - 3x^2 \\ p = 540 + 2x + 3x^2 \end{cases} \Rightarrow x = 10 \text{ and } \bar{p} = 860$$

$$p = 540 + 2x + 3x^2$$



The consumer surplus is

$$CS = \int_0^{10} (D(x) - \bar{p}) dx$$

$$= \int_0^{10} (1200 - 4x - 3x^2 - 860) dx$$

$$= (340x - 2x^2 - x^3) \Big|_0^{10}$$

$$= 2200$$

The producer surplus is

$$PS = \int_0^{10} (\bar{p} - S(x)) dx$$

$$= \int_0^{10} (860 - 540 - 2x - 3x^2) dx$$

$$= (320x - x^2 - x^3) \Big|_0^{10}$$

$$= 2100$$

$$\begin{cases} p = 1200 - 4x - 3x^2 \\ p = 540 + 2x + 3x^2 \end{cases} \Rightarrow x = 10 \text{ and } p = 860$$

$$p = 540 + 2x + 3x^2$$

The consumer surplus is

$$CS = \int_0^{10} (D(x) - p) dx$$

$$= \int_0^{10} (1200 - 4x - 3x^2 - 860) dx$$

$$= (340x - 2x^2 - x^3) \Big|_0^{10}$$

$$= 2200$$

The producer surplus is

$$PS = \int_0^{10} (p - S(x)) dx$$

$$= \int_0^{10} (860 - 540 - 2x - 3x^2) dx$$

$$= (320x - x^2 - x^3) \Big|_0^{10}$$

$$= 2100$$

$$\begin{cases} p = 1200 - 4x - 3x^2 \\ p = 540 + 2x + 3x^2 \end{cases} \Rightarrow x = 10, p = 860$$

$$p = 540 + 2x + 3x^2$$

The Consumer surplus is

$$CS = \int_0^{10} (D(x) - p) dx$$

$$= \int_0^{10} (1200 - 4x - 3x^2 - 860) dx$$

$$= (340x - 2x^2 - x^3) \Big|_0^{10}$$

$$= 2200$$

The producer surplus is

$$PS = \int_0^{10} (p - S(x)) dx$$

$$= \int_0^{10} (860 - 540 - 2x - 3x^2) dx$$

$$= (320x - x^2 - x^3) \Big|_0^{10}$$

$$= 2100$$

$$x = x_0 \cos t, y = y_0 \sin t$$



$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \cdot \frac{dt}{dx}$$

8. Let C be the curve parameterized by $x = t^2, y = t^3 - 9t$,

(1) Find the tangent line of the curve C at the point $(9, 0)$

(2) Find the points at which the curve C has horizontal tangent line.

(1) $m = \frac{dy}{dx} \Big|_{(9,0)} = \frac{3t^2 - 9}{2t}$

$\therefore (9, 0) = (t^2, t^3 - 9t) \Rightarrow t = 3 \text{ or } -3$

The slope of the tangent line is

$$\frac{dy}{dx} \Big|_{t=3} = \frac{3t^2 - 9}{2t} \Big|_{t=3} = 3$$

$$\frac{dy}{dx} \Big|_{t=-3} = \frac{3t^2 - 9}{2t} \Big|_{t=-3} = -3$$

$$\Rightarrow y - 0 = 3(x - 9)$$

$$y = 3x - 27$$

$$\Rightarrow y = -3(x - 9)$$

$$y = -3x + 27$$

(2) $\therefore \frac{dy}{dx} = 0 \therefore \frac{dy}{dt} = 0$ and $\frac{dx}{dt} \neq 0$

$$y = t^3 - 9t \Rightarrow \frac{dy}{dt} = 3t^2 - 9 = 0 \Rightarrow t = \pm\sqrt{3}$$

when $t = \sqrt{3}$,

$$x = 3, y = 3\sqrt{3} - 9\sqrt{3} = -6\sqrt{3}$$

when $t = -\sqrt{3}$

$$x = 3, y = -3\sqrt{3} + 9\sqrt{3} = 6\sqrt{3}$$

The point at horizontal tangent line are $(3, -6\sqrt{3})$ and $(3, 6\sqrt{3})$

$$m = \frac{dy}{dx} \Big|_{(9,0)} = \frac{3t^2 - 9}{2t}, \therefore (9, 0) = (t^2, t^3 - 9t) \Rightarrow t = 3 \text{ or } -3$$

The slope of the tangent line is

$$\frac{dy}{dx} \Big|_{t=3} = \frac{3t^2 - 9}{2t} \Big|_{t=3} = 3, \quad \frac{dy}{dx} \Big|_{t=-3} = \frac{3t^2 - 9}{2t} \Big|_{t=-3} = -3$$

$$\Rightarrow y = 3x - 27 \text{ and } y = -3x + 27$$

$$\frac{dy}{dx} = 0 \Rightarrow y = t^3 - 9t \Rightarrow \frac{dy}{dt} = 0 = 3t^2 - 9 \Rightarrow t = \pm\sqrt{3}$$

when $t = \sqrt{3}, x = 3, y = -6\sqrt{3}$

when $t = -\sqrt{3}, x = 3, y = 6\sqrt{3}$

The point are $(3, -6\sqrt{3})$ and $(3, 6\sqrt{3})$

$$m = \frac{dy}{dx} \Big|_{(9,0)} = \frac{3t^2 - 9}{2t}, \therefore (9, 0) = (t^2, t^3 - 9t) \Rightarrow t = 3 \text{ or } -3$$

The slope of the tangent line is

$$\frac{dy}{dx} \Big|_{t=3} = \frac{3t^2 - 9}{2t} \Big|_{t=3} = 3, \quad \frac{dy}{dx} \Big|_{t=-3} = \frac{3t^2 - 9}{2t} \Big|_{t=-3} = -3$$

$$\Rightarrow y = 3x - 27 \text{ and } y = -3x + 27$$

$$\frac{dy}{dx} = 0 \Rightarrow y = t^3 - 9t \Rightarrow \frac{dy}{dt} = 0 = 3t^2 - 9 \Rightarrow t = \pm\sqrt{3}$$

when $t = \sqrt{3}, x = 3, y = -6\sqrt{3}$

$t = -\sqrt{3}, x = 3, y = 6\sqrt{3}$

The point are $(3, -6\sqrt{3})$ and $(3, 6\sqrt{3})$